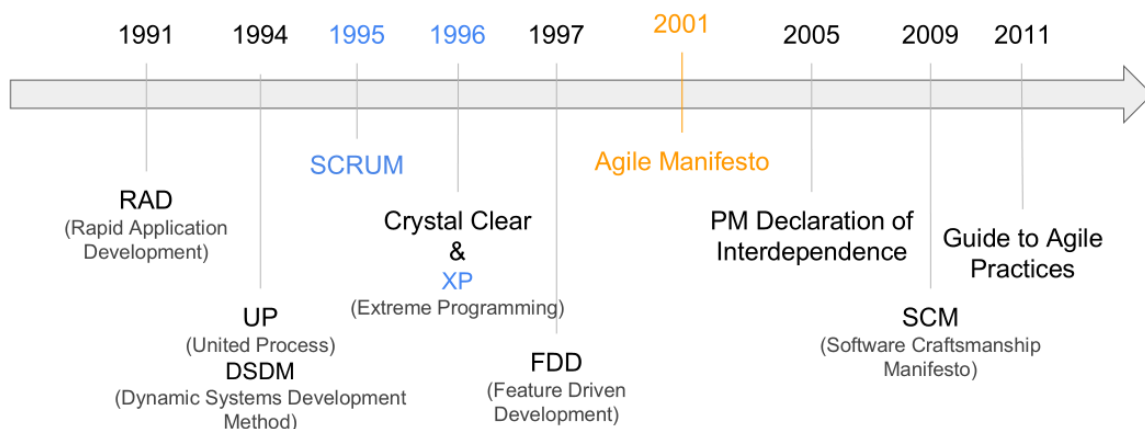


Agile Scrum

Ludovic

Quick Agile History

Quick Agile Story



Waterfall vs Agile

Waterfall: A linear and sequential approach where each phase must be completed before the next begins. It is often inflexible and can lead to challenges if requirements change.

Expression des besoins -> spécification -> conception générale -> conception détaillée -> réalisation -> tests unitaires -> tests d'intégration -> validation/recette -> maintenance

Agile: An iterative and incremental approach that emphasizes flexibility and customer collaboration. Agile allows changes to be made throughout the development process, accommodating evolving requirements.

Agile Methodologies

The Manifesto for Agile Software Development

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

Agile Software Development Principles

1. Customer satisfaction by early and continuous delivery of valuable software.

2. Welcome changing requirements, even late in development.
3. Deliver working software frequently (weeks rather than months).
4. Close, daily cooperation between business people and developers.
5. Projects are built around motivated individuals, who should be trusted.
6. Face-to-face conversation is the best form of communication (co-location).
7. Working software is the primary measure of progress.
8. Sustainable development, able to maintain a constant pace.
9. Continuous attention to technical excellence and good design.
10. Simplicity—the art of maximizing the amount of work not done—is essential.
11. Best architectures, requirements, and designs emerge from self-organizing teams.
12. Regularly, the team reflects on how to become more effective and adjusts accordingly.

For each Agile principle, take some time to reflect on them and explain their advantages:

- For the company
- For the team
- For the individual
- For the customer

1. Customer satisfaction by early and continuous delivery of valuable software.

- Company: Delivers value quickly and improves the company's image.
- Team: Feels their work is valuable; sense of accomplishment.
- Individual: Works on smaller tasks; frequent wins boost motivation.
- Customer: Receives a functional deliverable quickly.

2. Welcome changing requirements, even late in development.

- Company: Greater adaptability to customer demands.
- Team: Easier to handle changes and pivot when needed.
- Individual: Feels their work is meaningful and counted.
- Client: Can continuously adapt their needs.

3. Deliver working software frequently (weeks rather than months).

- Company: Increased responsiveness, more business opportunities.
- Team: Breaks the project into smaller pieces; less stress and discouragement.
- Individual: Lower risk of falling behind.
- Client: Can follow progress and iterate quickly.

4. Close, daily cooperation between business people and developers.

- Company: Better understanding of customer needs and an overall overview.
- Team: Ensures the team is on the right track.
- Individual: Direct feedback on their work.
- Client: Direct input on the solution, can propose changes, and ensure alignment.

5. Projects are built around motivated individuals, who should be trusted.

- Company: Greater productivity; less need to micromanage.

- Team: Better cohesion, higher motivation, increased productivity, fewer delays.
- Individual: Better work atmosphere and higher productivity.
- Client: Can trust the team and company, which increases client confidence.

6. Face-to-face conversation is the best form of communication (co-location).

- Company: Direct contact with staff; improved cohesion between management and development teams.
- Team: Improved communication and team cohesion.
- Individual: Better communication with others; feels valued and heard.
- Client: Better communication with the company; clients feel important and heard.

7. Working software is the primary measure of progress.

- Company: Ensures the product functions.
- Team: Verifies that work actually produces value.
- Individual: Reduced pressure from unnecessary artifacts.
- Client: Receives a functioning product.

8. Sustainable development, able to maintain a constant pace.

- Company: Maintains deadlines, reduces overtime costs, increases client trust.
- Team: Avoids crunch periods and respects deadlines.
- Individual: Prevents burnout and preserves long-term motivation.
- Client: Receives a functional product on time.

9. Continuous attention to technical excellence and good design.

- Company: Reputation for high-quality products and expertise.
- Team: Easier collaboration between team members.

- Individual: Cleaner work, less wasted time, greater satisfaction.





- Client: High-quality final product.

10. Simplicity—the art of maximizing the amount of work not done—is essential.

- Company: Less work, lower personnel needs, reduced costs, higher profit.
- Team: More effective and focused work.
- Individual: Lower workload and higher efficiency.
- Client: A simple and effective solution.

11. Best architectures, requirements, and designs emerge from self-organizing teams.

- Company: Less need for subcontracting.
- Team: A cohesive team that stays in a "closed loop".
- Individual: More desire to remain in the role, possibly beyond development.
- Client: Ends up with a well-designed product, increased trust, and better communication. :thumbs-up:

12. Regularly, the team reflects on how to become more effective and adjusts accordingly.

- Company: Saves time and money, collects constant feedback, and responds to needs.
- Team: Gains continuous feedback and improves processes.
- Individual: Becomes better, more confident, and feels heard.
- Client: Receives an increasingly better product.

Mayden Case Study

What challenges did the company face? What impact did it have on their business?

- The company faced challenges adapting to frequently changing client needs while using a Waterfall environment, which was not flexible enough. "Our best-laid plans were continually hijacked for short-priority developments. The end result was that we reached a point where we had started lots of things but finished very little."

There was an illusion of progress. Projects were assigned to individual contributors, taking months to finish. There was no team environment; each person was an expert in their own field. Some had large backlogs while others had insufficient work. In general, teams were bored, with low morale and motivation created by these individual silos.

What opportunity presented itself to the company? How was it embraced?

- The company had a new project with a brand-new technology. Management saw this as an opportunity to try a new approach and decided to use Agile Scrum. The company fully committed, providing training and coaching and allowing the team to self-organize.

What was the team reaction?

- Some team members already had Agile experience and introduced it to colleagues. They saw the benefits and were excited to try it.

What are the key elements to success in this transition?

1. The whole company must commit to the change, from top to bottom.
2. The team should be trained and coached in Agile.
3. Avoid falling into bad habits early on.

Can you identify 3 types of companies for which Scrum wouldn't apply as easily?

1. Construction companies, where designs cannot be changed once construction has started.
2. Companies in highly regulated environments (e.g., aerospace contractors) that must follow strict guidelines.
3. Organizations with extremely large teams where coordination and communication are difficult.

What would be a good definition of "Waterfall" according to you? Pick a company that you believe works in Waterfall and try to outline and design their processes.

- Waterfall is a linear and sequential approach to project management and software development, where each phase must be completed before the next begins. It is often inflexible and can lead to challenges if requirements change.

A company that works in Waterfall is a construction company. Their process would be:

1. Expression des besoins: The client expresses needs and requirements.
2. Spécification: Create a detailed specification, including design, materials, and budget.
3. Conception générale: Create general design, floor plans, elevations.
4. Conception détaillée: Create detailed structural and technical designs.
5. Réalisation: Begin construction.
6. Tests unitaires: Test individual building components (e.g., electrical systems, plumbing).
7. Tests d'intégration: Test integration of all components.
8. Validation/recette: Final inspection to ensure requirements and regulations are met.
9. Maintenance: Provide maintenance and repairs as needed.

Scrum Framework

Introduction to Scrum

Framework: A set of guidelines or a structure that helps organize and guide the work of a particular system or process.

Defined in The Scrum Guide by Ken Schwaber and Jeff Sutherland.

A better way for teams to collaborate when solving complex problems.

Scrum is defined by its Roles, Events, and Artifacts within a bounded framework, as well as its three pillars: Transparency, Inspection, and Adaptation; and five values: Commitment, Courage, Focus, Openness, and Respect.

Scrum Roles

- Scrum Master:
 - Facilitates the Scrum process.
 - Ensures that the Scrum process is followed.
 - Helps the team stay focused on their goals.
 - Removes obstacles that prevent progress.
 - Ensures the team follows Scrum values and principles.
- Product Owner:
 - Represents the interests of stakeholders.
 - Defines the goals and priorities of the project.
 - Creates and maintains the product backlog.
 - Communicates the vision and goals of the project to the team.
 - Ensures the team is working on the most important tasks.
- Development Team:
 - Delivers working software at the end of each sprint.
 - Self-organizing and self-managing.
 - Cross-functional members with the skills needed to complete the work.
 - Responsible for completing all tasks required to deliver a working product.

Scrum Ceremonies

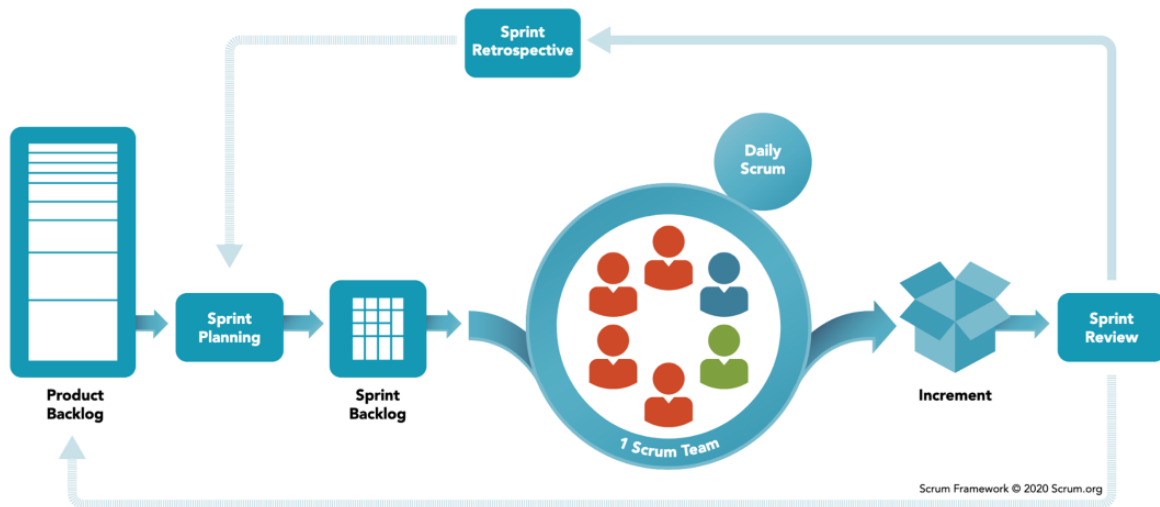
- Sprint Planning:
 - What will we do?
- Daily Scrum:
 - What are we doing?
- Sprint Review:
 - What did we do?
- Sprint Retrospective:
 - How can we improve? How did we do?

Scrum Artifacts

- Product Backlog
 - Product Goal — the target the team aims to achieve
- Sprint Backlog
 - Sprint Goal — the target for the sprint
- Increment

- Definition of Done — the description of what it takes for an increment to be considered "done"

Overview



Product & Planning

Product Ownership

"Defining, prioritizing, and communicating the features and capabilities of a product to ensure that it meets the needs and expectations of customers and stakeholders."

Critical in agile development because it helps ensure the product is delivered on time, to high quality, and with maximum value to the customer.

Role of a Product Owner

The Product Owner is responsible for representing the customer's interests and setting the strategic direction for the product. This includes:

- defining the product vision, roadmap, and release plan,
- prioritizing the product backlog to ensure the most valuable features are delivered first.

Responsibilities of a Product Owner

- Define and communicate the product vision.
- Prioritize and manage the product backlog.
- Collaborate with cross-functional teams.
- Communicate progress and updates to stakeholders.

Product Owner vs Project Manager

- Product Owners define product vision and strategy, and prioritize the product backlog.

- Project Managers plan and execute projects, manage resources and budgets to ensure delivery on time and within scope.
- Both roles require leadership, communication, and problem-solving skills, but they focus on different responsibilities.

Product Vision & Strategy

Defining the Product Vision

"A product vision provides a long-term direction for the product and helps align stakeholders and cross-functional teams around a common goal."

- Steps for defining the product vision:
 - Identify the target market and customer needs.
 - Determine the value proposition.
 - Define the product vision statement.

Product Roadmap

A product roadmap is a high-level overview of the features and capabilities that will be included in the product over time. It aligns stakeholders and teams around a common vision and provides direction for development.

Steps for developing a product roadmap:

- Define the long-term goals and objectives for the product: What do you want to achieve, and how will it benefit customers and the business?
- Break down the goals into smaller, manageable chunks that can be accomplished over shorter time periods.
- Prioritize the objectives: determine which objectives are most important and should be tackled first.

Prioritization

Prioritization is critical because it ensures the most valuable features are delivered first and resources are used efficiently.

Techniques for prioritizing the product backlog:

- Value vs. effort matrix: Plot each feature based on its value to the customer and the effort required.
- Kano analysis: Group features by how they affect customer satisfaction; prioritize high-impact features.
- MoSCoW method: Categorize features as "Must have," "Should have," "Could have," or "Won't have."

User Stories & Prioritization

What are User Stories?

User stories are short, simple descriptions of a feature from the user's perspective. They are used in agile to capture requirements and guide development.

Usual format: As a [user], I want [feature], so that [benefit].

Acceptance criteria: Specific, measurable criteria that must be met for the user story to be complete.

Benefits of User Stories

- Focus on user needs and goals rather than technical details.
- Facilitate collaboration between product owners, developers, and stakeholders.
- Provide a flexible way to capture and prioritize requirements.

Tips for writing User Stories

- Make the user story clear and concise.
- Ensure it has a clear value proposition for the user.
- Make acceptance criteria specific and measurable.
- Use examples and scenarios to illustrate the story.
- Collaborate with cross-functional teams to ensure feasibility.

Definition of Done

A set of specific criteria that must be met for a user story or feature to be considered complete.

Key components that may be included in the Definition of Done:

- Code has been reviewed and is of high quality.
- Unit tests and automated tests have been created and run successfully.
- User documentation has been created and reviewed.
- Feature has been tested by QA and issues resolved.
- Feature has been reviewed by the Product Owner and stakeholders and approved for release.

Minimum Viable Product (MVP)

- The MVP is a product with just enough features to satisfy early customers and gather feedback for future development.
- It is a way to test a product hypothesis with minimal effort and resources.
- The MVP allows rapid prototyping and iteration based on customer feedback.
- It minimizes risk by gathering data before committing significant resources.
- The MVP focuses development on the most necessary features instead of building everything at once.

What to include

- Identify the core value proposition.
- Prioritize based on customer needs.
- Consider feasibility.
- Balance risk and reward.
- Gather feedback.

Kanban & General Considerations

Kanban

Kanban is a method for managing and improving workflows, developed by Toyota in the 1950s. It is based on just-in-time (JIT) production principles.

Aims to increase efficiency and reduce waste by:

- Visualizing work.
- Limiting work-in-progress (WIP).
- Continuously improving processes.

Visualizing Work

Kanban board: A visual representation of the work process, showing the status of each task as it moves through the workflow.

- Work-in-progress (WIP) limits: Maximum number of tasks allowed in each stage to prevent overload and bottlenecks.
- Pull system: Start work only when there is capacity to complete it.
- Continuous improvement: Regularly review and optimize the workflow to identify and address bottlenecks.

All in all: Agile best practices

Lean

Lean is a business philosophy that focuses on maximizing customer value while minimizing waste. It emphasizes continuous improvement, respect for people, and delivering customer value.

Key Components of Lean

- Value stream mapping: Identify and analyze the flow of value from the customer's perspective to eliminate waste.
- Continuous improvement: Regularly identify and address opportunities to increase efficiency.
- Respect for people: Empower team members and recognize their value.
- Customer focus: Use customer feedback to guide decisions and improvements.

Lean Principles

- Define value from the customer's perspective.
- Identify and eliminate waste.
- Create flow in the value stream.
- Use pull systems to trigger production.
- Strive for perfection.

Lean Tools and Techniques

- Value stream mapping
- Six Sigma
- Kanban

- 5S
- Standardized work
- Visual management
- Root cause analysis
- Kaizen events

Aspect	Scrum	Nexus
Objectif	Gérer une seule équipe Scrum	Coordonner plusieurs équipes Scrum
Valeurs	Identiques	Identiques
Rôles	PO, SM, Devs	PO, SM, Devs + Nexus Integration Team
Événements	5 événements Scrum	5 événements Scrum + variantes Nexus
Artefacts	Product Backlog, Sprint Backlog, Increment	Les mêmes, mais à l'échelle Nexus
Principe	Empirisme (Inspection, Adaptation, Transparence)	Idem
Incrément	Produit livrable d'une équipe	Produit intégré de plusieurs équipes

Extreme Programming (XP)

Key components:

- Pair programming
- Test-driven development (TDD)
- Continuous integration
- Continuous delivery
- Collective ownership
- Customer collaboration

Test-Driven Development (TDD)

- TDD is a development methodology in which developers write automated tests for new code before writing the code itself.
- Tests define desired behavior and requirements and are run after the code is written to ensure it works as expected.
- The TDD cycle: write a test, run the test and confirm it fails, write code to make the test pass, run tests again to ensure they pass, and refactor as needed.
- TDD encourages better design, fewer bugs, and a reliable test suite.